From Erasure to CRUD: A Unified Information-Thermodynamic Principle

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Abstract

Landauer's principle is usually formulated as a lower bound on the thermodynamic cost of erasure. In this work, we argue that erasure is only one special case of a broader information-thermodynamic principle that naturally extends to create, read, update, and delete (CRUD) operations. We formulate this principle in terms of nonequilibrium free energy, such that the average work satisfies

$$\langle W \rangle \;\geq\; \Delta F_{\mathrm{eq}} + kT\,\Delta D_{\mathrm{KL}}^{\mathrm{full}} \;\equiv\; \Delta F_{\mathrm{total}},$$

where ΔF_{eq} is the equilibrium free-energy difference between endpoint Hamiltonians and $\Delta D_{\mathrm{KL}}^{\mathrm{full}}$ is the change in Kullback–Leibler divergence between the actual state distribution and the corresponding equilibrium distribution. We evaluate this principle in a unified stochastic memory model based on a driven double-well potential and a coupled readout meter. Within this framework, Delete and overwrite-like Update exhibit information free-energy changes close to the Landauer scale $kT\ln 2$, whereas Create behaves as a write-like operation whose state-function contribution is near zero and is therefore not constrained by the erasure form of Landauer's bound. Read is captured as a measurement process with $\Delta F_{\mathrm{eq}} = 0$ but positive dissipative cost associated with correlation generation. The same code base generates all CRUD figures and data tables. These results support the view that Landauer's principle should be understood not as a law for deletion alone, but as the deletion-specific limit of a broader CRUD information thermodynamics.

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1. Introduction

Digital information processing is commonly organized in terms of four elementary actions: create, read, update, and delete. Among these, Landauer's principle [1] is traditionally associated with deletion, stating that logically irreversible erasure carries a minimum thermodynamic cost. This perspective has been extraordinarily influential [2, 3], but it leaves open a broader question: if information is a physical resource, should thermodynamic constraints apply only to deletion, or should they govern the full CRUD structure of information processing?

The present work addresses this question by treating CRUD operations as transformations of probability distributions over physical memory states. From this viewpoint, deletion is not the only relevant operation. Writing, overwriting, and measurement can also reshape state distributions, change information content, or generate correlations. What distinguishes deletion is not that it is the sole information-thermodynamic operation, but rather that it is a particular macro-irreversible case in which the information free-energy change is close to the familiar $kT \ln 2$.

Recent developments in the thermodynamics of computation have reinforced this perspective. Within the stochastic thermodynamic framework reviewed by Seifert [4], Boyd, Mandal, and Crutchfield [5] showed that modular, localized information processing can generate unavoidable dissipation beyond the Landauer bound. Wolpert and coauthors [6] argued that stochastic thermodynamics provides the appropriate framework for relating energetic cost to computational structure and physical constraints, rather than relying on Landauer's bound alone. Manzano *et al.* [7] extended the thermodynamic theory of computation to processes with absolute irreversibility, unidirectional transitions, and stochastic computation times. In a related direction, Hsieh [8] established an analytical correspondence between classical information transmission and energy extraction, uncovering a dynamical version of Landauer's principle. These developments are placed in broader context by a recent comprehensive review [9] covering finite-time protocols, non-Markovian dynamics, and experimental validation across multiple platforms. Together, these lines of research suggest that the thermodynamic

cost of computation is shaped not only by erasure, but also by architecture, modularity, and the full probabilistic structure of information-bearing states. On the measurement side, Sagawa and Ueda [10, 11] established the thermodynamic framework for feedback-controlled systems and derived the generalized second law incorporating mutual information, which provides the theoretical basis for our treatment of the Read operation. The recent experimental probing of Landauer's principle in quantum many-body systems [12] further demonstrates that such principles extend well beyond classical single-bit models.

Our central claim is that Landauer's principle should be understood as a special case of a more general information-thermodynamic inequality. We formulate that inequality using nonequilibrium free energy and test it numerically in a single stochastic memory model for Create, Read, Update, and Delete. In this model, Delete and overwrite-like Update exhibit an information free-energy change close to $kT \ln 2$, whereas Create behaves as a write-like operation whose free-energy change is near zero and is therefore not $kT \ln 2$-limited. Read is treated separately as a measurement process that creates correlation rather than logical compression, in alignment with the Sagawa–Ueda framework for measurement thermodynamics [10]. This unified picture allows us to place all four CRUD operations within one thermodynamic accounting framework, providing a concrete state-distribution realization of the broader post-Landauer program advocated in the recent literature.

2. Framework

We consider a physical memory described by a control-dependent Hamiltonian $H(x;\lambda)$, where x is the microscopic state and λ denotes externally driven protocol parameters. For a nonequilibrium state distribution $p(x)$, we define the nonequilibrium free energy [4, 13] as

$$F[p;\lambda] = F_{\mathrm{eq}}(\lambda) + kT\, D_{\mathrm{KL}}\big(p(x) \,\|\, \pi_\lambda(x)\big), \tag{1}$$

where $\pi_\lambda(x) \propto e^{-\beta H(x;\lambda)}$ is the equilibrium distribution and $F_{\mathrm{eq}}(\lambda)$ is the corresponding equilibrium free energy. For isothermal driving without feedback, the average work satisfies [14]

$$\langle W \rangle \;\geq\; \Delta F_{\mathrm{eq}} + kT\big[D_{\mathrm{KL}}(p_f \| \pi_f) - D_{\mathrm{KL}}(p_i \| \pi_i)\big] \;\equiv\; \Delta F_{\mathrm{total}}. \tag{2}$$

We take this inequality as the generalized CRUD form of Landauer's principle. In this formulation, deletion is not singled out a priori. Instead, every operation is assigned a thermodynamic lower bound through the same state-function difference ΔF_{total}.

To clarify how logical information enters this bound, we decompose the full Kullback–Leibler divergence into a basin-level macro contribution and an intra-basin contribution. Let the logical state be the sign of x, with left and right wells corresponding to $m \in \{L, R\}$. Then

$$D_{\text{KL}}^{\text{full}}(p\|\pi) = D_{\text{KL}}^{\text{macro}}(p_m\|\pi_m) + \sum_{m\in\{L,R\}} p_m \, D_{\text{KL}}\big(p(x|m) \,\|\, \pi(x|m)\big). \tag{3}$$

The first term measures logical compression or expansion, whereas the second term quantifies intra-well distortion relative to conditional equilibrium. In the symmetric erase case, $D_{\text{KL}}^{\text{macro}}$ reduces to the usual Landauer form $\ln 2 - h(\varepsilon)$, where ε is the final logical error rate.

Within this language, the CRUD operations are interpreted as follows. **Create** is modeled as $\text{blank}_0 \to 1$, i.e., a write-like transfer from a known state to another known state. **Update** is modeled as unknown $\to 1$ overwrite, which contains an erase-like component because the previous unknown value is not retained. **Delete** is modeled as unknown $\to \text{blank}_0$ and therefore represents the standard Landauer-like case. **Read** is different: it ideally leaves the system Hamiltonian unchanged and instead creates correlation between the system and a meter. For Read we therefore use $\Delta F_{\text{eq}} = 0$ and evaluate the dissipative work together with the generated mutual information $I(X : Y)$, which must satisfy the measurement bound [10, 11]

$$W_{\text{diss}}^{\text{read}} \;\geq\; -kT\, I(X : Y). \tag{4}$$

This provides a unified picture in which Delete is the $kT \ln 2$ special case, while Create, Update, and Read are governed by the same broader information-thermodynamic structure.

3. Method

We use a one-dimensional overdamped Langevin memory model with double-well potential

$$V(x; b, c) = x^4 - bx^2 - cx, \tag{5}$$

where b controls the barrier height and c controls the left–right bias. The state variable $x < 0$ corresponds to logical 0 and $x > 0$ to logical 1. Time evolution is generated by Euler–Maruyama integration of the overdamped Langevin equation at fixed temperature T. Work is accumulated solely through discrete protocol changes in the control parameters (b, c),

$$\delta W = V(x; \lambda_{i+1}) - V(x; \lambda_i). \tag{6}$$

Create, Update, and Delete are implemented by a four-stage protocol: barrier lowering, bias application, barrier raising, and optional unbiasing at high barrier. In the present paper we use the "unbias" endpoint mode, so that the initial and final Hamiltonians coincide and the endpoint equilibrium free-energy difference is approximately zero. Initial states are drawn canonically from either the full equilibrium distribution or the conditional equilibrium distribution within a selected basin. Specifically, Create starts from the left basin (blank_0), Update starts from the full equilibrium mixture (unknown old value), and Delete starts from the full equilibrium mixture and targets the left basin. For Create, we calibrate the write bias in two steps: first by solving analytically for the low-barrier equilibrium right-well probability, and second by a small endpoint micro-correction so that the final logical error is close to the prescribed target $\varepsilon_{\text{write}} = 0.01$.

The information free-energy term is evaluated from the endpoint distributions. The full contribution is computed from a histogram approximation to $D_{\text{KL}}(p\|\pi)$, while the macro/intra decomposition is obtained by splitting the state space into left and right basins and applying the chain rule of relative entropy. This decomposition allows us to separate logical compression from intra-basin nonequilibrium deformation. All core results reported below use $T = 1$, $b_{\text{hi}} = 4.0$, $b_{\text{lo}} = 0.2$, $n_\lambda = 2000$, and relax $= 5$ Langevin steps per protocol increment, while the relax scan explores $1, 2, 5, 10, 20$ steps per increment.

Read is treated as a coupled system-meter dynamics with total potential

$$V_{\text{tot}}(x, y) = V(x; b, c) + \tfrac{1}{2}k_m y^2 - g\,xy, \tag{7}$$

where y is the meter coordinate, k_m is the meter spring constant, and g is a time-dependent coupling. The coupling is ramped $0 \to g_{\max} \to 0$, so the Hamiltonian returns to its initial form and $\Delta F_{\text{eq}} = 0$. Mutual information is estimated by a two-dimensional histogram of (x, y), and measurement accuracy is estimated from the sign agreement between system and meter.

All simulations were implemented in a single JAX-based code running on GPU and exported directly to CSV and publication figures from the same code path.

4. Results

Figure 1 summarizes the thermodynamic decomposition of Create, Read, Update, and Delete within a single computational framework. In the core configuration, Create yields $\langle W \rangle = 1.6067$, $\Delta F_{\text{total}} = -0.0373$, and $W_{\text{diss}} = 1.6440$, with final logical error $\varepsilon = 0.0078$. The small magnitude of ΔF_{total} shows that Create is not constrained by the $kT \ln 2$ Landauer limit in the same way as erasure; rather, it behaves as a write-like operation whose state-function contribution is near zero and whose current work budget is dominated by protocol dissipation.

Update, modeled as unknown $\to$ 1 overwrite, gives $\langle W \rangle = 1.5989$, $\Delta F_{\text{total}} = 0.6561$, and $W_{\text{diss}} = 0.9428$, with $\varepsilon = 0.0079$. The information free-energy change is dominated by the macro term ($\Delta F_{\text{macro}} = 0.6470$) and lies close to $kT \ln 2$, indicating that overwrite without retention of the old value contains an erase-like component. Delete, modeled as unknown $\to$ blank$_0$, gives $\langle W \rangle = 1.6203$, $\Delta F_{\text{total}} = 0.6745$, and $W_{\text{diss}} = 0.9458$, with $\varepsilon = 0.0048$. Again the macro term dominates ($\Delta F_{\text{macro}} = 0.6627$), and the total information free-energy change is close to the Landauer value $kT \ln 2 = 0.6931$. These two cases show that the Landauer-scale cost emerges naturally whenever the operation compresses an initially unknown logical state into a specified target state.

Read displays a different thermodynamic structure. Because the coupling protocol returns the Hamiltonian to its initial form, $\Delta F_{\text{eq}} = 0$ by construction. Nevertheless, the measurement incurs positive dissipative work, $\langle W \rangle = W_{\text{diss}} = 1.6721$, while generating mutual information $I(X : Y) = 0.4184$ bits with accuracy 79.13%. The measured dissipative work remains above the Sagawa–Ueda bound $-kT\, I$, confirming that readout is thermodynamically nontrivial even when no logical compression occurs [10]. This distinguishes Read from Delete while keeping it inside the same information-thermodynamic picture.

The relax scan further separates state-function contributions from protocol irreversibility. For Delete, W_{diss} decreases from 3.3886 at relax = 1 to 0.3208 at relax = 20, while ΔF_{total} stays in the 0.61–0.67 range. Update behaves similarly, with W_{diss} decreasing from 3.6315 to 0.3223. Create also

shows a systematic reduction of W_{diss}, from 5.1134 to 1.0099, while its $\Delta F_{\mathrm{total}}$ remains close to zero and may become slightly negative. Taken together, these trends show that protocol dissipation and information free-energy are distinct quantities: slower driving reduces W_{diss}, whereas the logical character of the operation is encoded in $\Delta F_{\mathrm{total}}$. This is precisely the sense in which Landauer's principle can be generalized from Delete alone to a CRUD-wide information-thermodynamic framework.

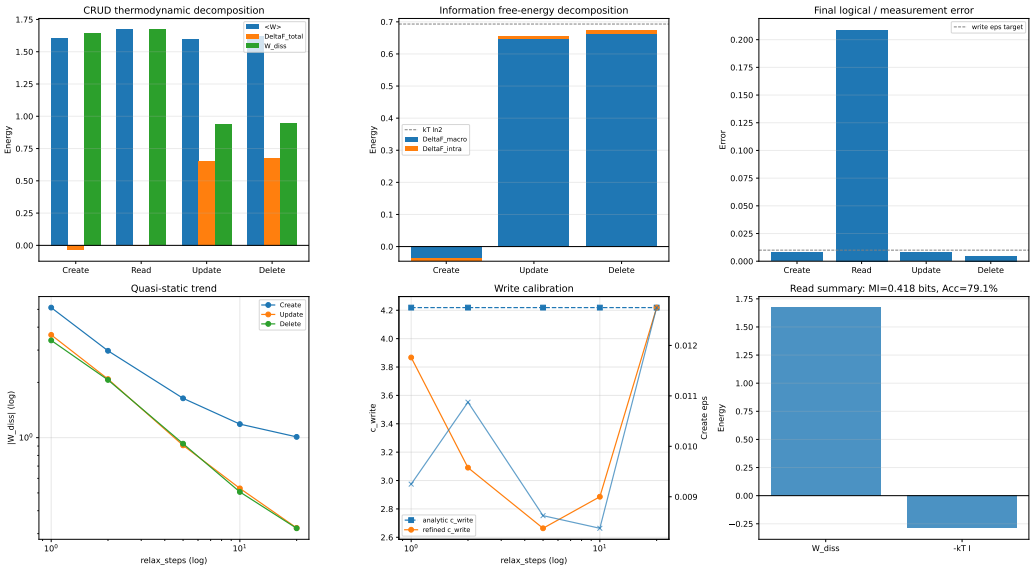

Figure 1: Integrated CRUD information thermodynamics in a single stochastic memory model. (a) Mean work $\langle W \rangle$, total nonequilibrium free-energy change $\Delta F_{\mathrm{total}} = \Delta F_{\mathrm{eq}} + kT\,\Delta D_{\mathrm{KL}}^{\mathrm{full}}$, and dissipative work $W_{\mathrm{diss}} = \langle W \rangle - \Delta F_{\mathrm{total}}$ for Create, Read, Update, and Delete in the core configuration ($T = 1$, $b_{\mathrm{hi}} = 4.0$, $b_{\mathrm{lo}} = 0.2$, $n_\lambda = 2000$, relax $= 5$). (b) Decomposition of the information free-energy change into macro and intra components for Create, Update, and Delete. The dashed line marks $kT \ln 2$. (c) Final task error for Create, Update, and Delete, together with the Read measurement error. (d) Relax-scan of $|W_{\mathrm{diss}}|$ for Create, Update, and Delete, showing the reduction of dissipative cost under slower driving. (e) Analytic plus endpoint-corrected Write calibration across relax values, used to keep the final Create error near $\varepsilon_{\mathrm{write}} = 0.01$. (f) Read panel comparing measured dissipative work to the Sagawa–Ueda lower bound $-kT\,I(X:Y)$, with $I(X:Y)$ estimated from the final joint system-meter distribution. The figure shows that Delete and overwrite-like Update fall near the Landauer scale, while Create and Read are governed by the broader CRUD information-thermodynamic bound rather than by $kT \ln 2$ alone.

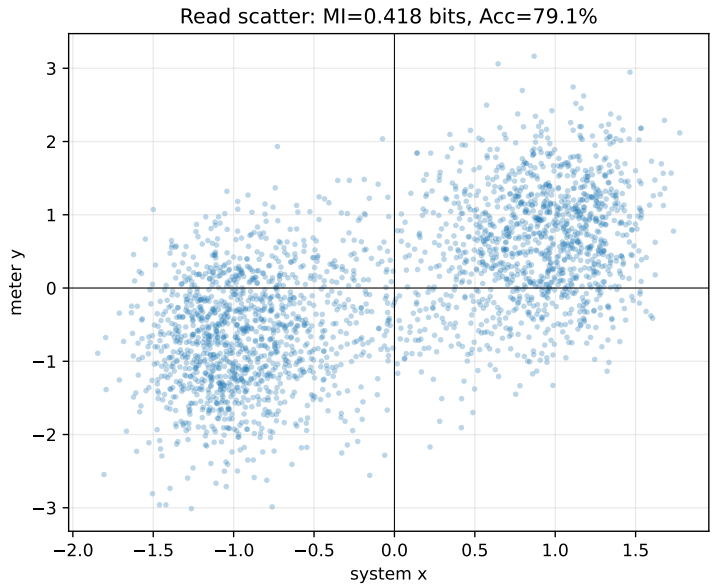

Figure 2: Final system-meter scatter for the Read operation. Scatter plot of the final system state x and meter output y for the Read protocol ($g_{\max} = 2.0$, $k_m = 2.0$, $T = 1$). The correlation between x and y yields mutual information $I(X : Y) = 0.4184$ bits and measurement accuracy 79.13%, while the Hamiltonian cycle enforces $\Delta F_{\text{eq}} = 0$. The energetic cost of Read therefore appears as dissipative work associated with correlation generation rather than logical compression.

5. Discussion

5.1. CRUD as a generalized Landauer framework

Landauer's principle is usually introduced as a statement about erasure. Our results suggest a broader interpretation: erasure is not the only information-processing operation with thermodynamic structure, but rather a special case within a more general CRUD information thermodynamics. In our framework, all operations are evaluated through the same state-function difference,

$$\Delta F_{\text{total}} = \Delta F_{\text{eq}} + kT\,\Delta D_{\text{KL}}^{\text{full}}, \tag{8}$$

so that the energetic cost of computation is governed not only by equilibrium free-energy changes, but also by the nonequilibrium information content of the initial and final state distributions. Under this view, Delete appears as the familiar Landauer-like case because it compresses an initially unknown logical state into a specified target state, while Create and Update need not

be constrained by the same $kT \ln 2$ form. Read occupies a distinct branch in which the endpoint Hamiltonian can remain unchanged while dissipative work is still required to generate correlations between system and meter. This provides a natural CRUD-level extension of Landauer's original insight.

5.2. Relation to modularity dissipation and mismatch cost

This positioning aligns with a broader recent shift in the thermodynamics of computation. In particular, Boyd, Mandal, and Crutchfield [5] showed that modular, localized information processing can incur dissipation above and beyond the Landauer cost associated with single-gate logical irreversibility, thereby making clear that focusing on erasure alone is insufficient for characterizing the full thermodynamic burden of computation. Our CRUD formulation is compatible with this perspective: the quantity W_{diss} in our simulations can be interpreted as the residual protocol-dependent or architectural overhead beyond the state-function contribution ΔF_{total}. In that sense, the present work may be viewed as a concrete stochastic-memory realization of the broader claim that computational thermodynamics cannot be reduced to deletion costs alone.

The same perspective is reinforced by the PNAS article of Wolpert *et al.* [6], which argues that the energetic cost of computation should be analyzed together with computational structure and physical constraints, rather than through Landauer's bound in isolation. Our CRUD model operationalizes that idea in an explicit one-bit stochastic memory system: Delete and overwrite-like Update display Landauer-scale macro compression, whereas Create behaves as a write-like operation whose ΔF_{total} is close to zero and whose energetic cost is therefore dominated by dissipation rather than by an irreducible $kT \ln 2$ contribution.

A related line of work concerns mismatch cost. Yadav, Caravelli, and Wolpert [15] argued that mismatch cost can furnish lower bounds on entropy production for Boolean circuits and random-access stored-program machines, and proposed thermodynamic cost as a third computational resource alongside time and space. Because that work remains a preprint, its status differs from that of the peer-reviewed articles discussed above; nonetheless, it is highly relevant conceptually. Our CRUD framework is complementary to that agenda. Whereas mismatch-cost analyses emphasize the gap between machine design and actual usage, the present model focuses on a state-distribution accounting that is directly computable for Create, Read, Update, and Delete within a single code base. One promising next step is

to connect these views explicitly by interpreting our residual W_{diss} as a decomposition into protocol-speed effects, modularity costs, and mismatch-like costs associated with using a given physical memory operation for a target computational task.

5.3. What is special about Delete, and what is not

A natural question arising from the CRUD framework is why Delete has historically occupied a privileged position. Within the present model, the answer is clear: Delete is the operation in which $\Delta D_{\mathrm{KL}}^{\mathrm{macro}}$ is maximal, because the initial distribution is maximally uncertain (full equilibrium mixture) and the final distribution is concentrated in a single basin. This is the precise condition under which the information free-energy change approaches $kT \ln 2$. Update shares this property partially, because it also starts from an unknown state, but its endpoint bias structure may differ depending on the target value.

Create, by contrast, starts from a known state (blank_0 in the left well) and targets another known state (logical 1 in the right well). Because both endpoints are macroscopically specified, the macro KL term $\Delta D_{\mathrm{KL}}^{\mathrm{macro}}$ can be close to zero or even negative, and the $kT \ln 2$ form does not apply. This observation is fully consistent with the well-known result that a deterministic state-to-state transformation can in principle be performed at zero cost [2, 1]: Create is precisely such an operation. Our contribution is to show that this distinction emerges naturally and quantitatively within the unified CRUD free-energy accounting, rather than requiring a separate argument.

Read is different again. Because the system Hamiltonian returns to its initial form at the end of the measurement protocol, $\Delta F_{\mathrm{eq}} = 0$ and the macro KL divergence of the system alone need not change. The thermodynamic cost instead appears as dissipative work required to establish correlation between system and meter, governed by the Sagawa–Ueda measurement bound [10, 11]. This confirms that Read is thermodynamically nontrivial even though it involves no logical compression.

5.4. Limitations and experimental outlook

The present study has several limitations. First, our results are model-based and numerically verified in a stochastic double-well memory rather than experimentally validated on production hardware. Second, the protocol families used here are not thermodynamically optimal, as shown by the nonzero dissipation that remains even after partial quasi-static refinement.

Optimal-transport-based protocols [16, 17] could reduce W_{diss} further and sharpen the distinction between state-function and protocol contributions. Third, Read is treated through a system-meter coupling model and therefore enters the CRUD framework through correlation generation rather than through logical compression. These limitations do not undermine the central claim, but they do delimit its scope: what is established here is a theoretically explicit and numerically tested CRUD extension of Landauer's principle within a specified stochastic memory model, not yet a universal empirical law for all computing substrates.

Experimental tests of the CRUD framework are in principle accessible using existing platforms. Colloidal particle systems of the type used by Bérut *et al.* [3] could be adapted to implement all four CRUD operations by varying the initial state preparation and protocol endpoint. Single-electron devices [18] provide another natural test bed. Such experiments would allow direct measurement of the state-function contribution $\Delta F_{\mathrm{total}}$ and its decomposition into macro and intra-basin terms, providing an empirical test of the predictions reported here.

6. Conclusion

We have proposed a CRUD extension of Landauer's principle by formulating information processing in terms of nonequilibrium free-energy changes rather than erasure alone. The resulting state-function bound,

$$\langle W \rangle \;\geq\; \Delta F_{\mathrm{eq}} + kT\,\Delta D_{\mathrm{KL}}^{\mathrm{full}}, \tag{9}$$

provides a common thermodynamic language for Create, Read, Update, and Delete. Within our stochastic memory model, Delete and overwrite-like Update appear as macro-irreversible operations with information free-energy changes near $kT\ln 2$, while Create behaves as a write-like operation whose thermodynamic lower bound is generally not of Landauer form. Read, in turn, is naturally interpreted as a measurement process whose cost is associated with correlation generation rather than logical compression.

These results suggest a conceptual reframing: Landauer's principle may be most naturally understood as the deletion-specific limit of a broader CRUD information thermodynamics. From this perspective, the relevant invariant is not logical irreversibility alone, but the full change in nonequilibrium information free energy carried by a physical state distribution. The

single-code numerical framework developed here demonstrates that this general principle can be instantiated and tested across all four CRUD operations within one consistent model.

The present work remains a model-based and numerically validated study rather than an experimental validation on production computing hardware. Nevertheless, it provides a concrete and unified foundation for future studies of information-processing thermodynamics in realistic devices, modular computational architectures, and engineered memory systems. In that sense, the CRUD extension offered here is not a replacement for Landauer's principle, but its natural generalization.

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